

MURISPHERE SOP TUTORIAL

Workflow Training Guide: Technician + Facility Manager

This tutorial is organized by real routine workflows in a research vivarium. Follow sections in order for onboarding, or jump by role and task.

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1) Quick Start and Prerequisites

Core principle: **cage-level speed and accuracy first**. Use phone scan workflows in the room, then manager workflows for oversight and planning.

Technician

PI/Manager

Admin

1. Sign in from phone or tablet.
2. Set **Scan Base URL** to a phone-reachable host (LAN IP or domain).
3. Print cage cards at 100% scale.
4. Verify one sample scan before starting room rounds.

2) Mouse Technician Daily Workflows

2.1 Start-of-Shift Room Entry

1. Open active alerts; identify **high-severity** cages first.
2. Review due/overdue tasks for assigned rooms.
3. Confirm protocol blocks are cleared before edits.

2.2 Phone Scan of Printed Card (Required)

1. Open phone camera and scan printed QR (not the 1D barcode strip).
2. Tap link banner to open cage page in browser.
3. Verify cage ID and location before edits.
4. Save count/status/note updates.

Never edit unless scanned cage ID matches printed card.

2.3 Cage Census and Reconciliation

1. Start census session for room.
2. Scan each cage and confirm observed M/F counts.
3. Complete session and record mismatch notes.

2.4 Breeding Operations

1. Create/manage breeding pairs (sire + dam + cage).
2. Record timed mating / plug check / weaning events.
3. Review pair productivity and retire non-productive pairs.

2.5 Health, Vet, Mortality, and Euthanasia

1. Run health rounds and log findings.
2. Open vet case and treatment plan as needed.
3. Record mortality and flag necropsy if required.
4. Record euthanasia events with method/reason/disposition.

2.6 Tagging and Sample Handling

1. Apply animal tags (ear tag/chip/tube/well).
2. Create sample record and sample code.
3. Advance status: collected → shipped → received → resulted.
4. Confirm chain-of-custody events are logged.

3) Facility Manager Daily Workflows

3.1 Morning Command Review

1. Review alert feed and category distribution.
2. Acknowledge or escalate unresolved alerts.
3. Dispatch notifications for urgent items.

3.2 Capacity and Room Operations

1. Check room/rack capacity and quota utilization.
2. Review cage-space projection and consolidation suggestions.
3. Prioritize cage wash and room transfer actions.

3.3 Recommendation Decisions

1. Generate recommendations (e.g., consolidation, supply gap).
2. Decision each item: accept, adjust, ignore, complete.
3. Track decision outcomes for continuous improvement.

3.4 Planner Scenario Management

1. Create scenario (target animals, needed-by date, cage constraints).
2. Attach projects and required animal counts.
3. Evaluate deficit/risk and review generated plan snapshots.

4) Regulatory and Compliance Workflows

- Protocol expiration monitoring and hard-stop edit enforcement.
- Deviation logging with CAPA and closure status.
- Quarantine intake, release, and overdue monitoring.
- Qualification-expiry monitoring for assigned procedures.
- E-signature capture and attachment evidence for inspections.
- Audit trail review for who/what/when changes.

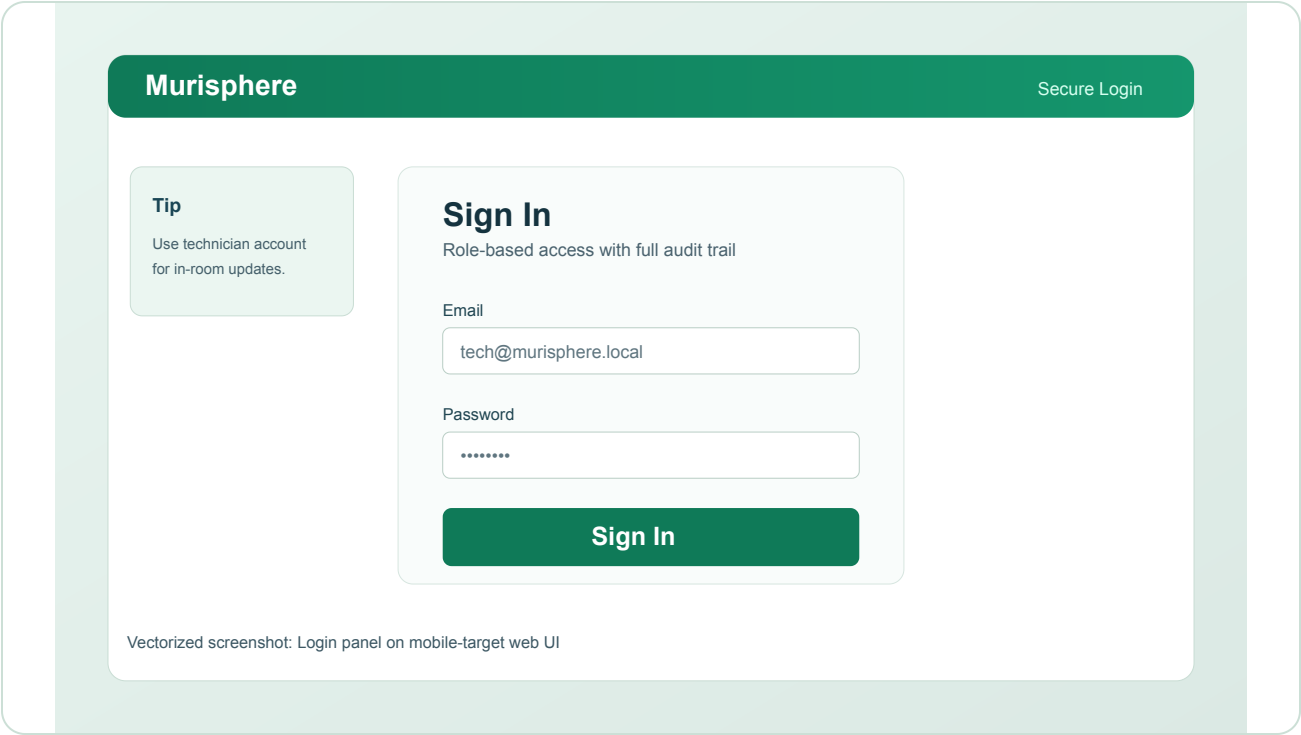
5) Research Support Workflows

- Pedigree exploration for multi-generation lineage decisions.
- Genotyping order submission and callback result ingestion.
- Mendelian ratio and genotype-deviation alert review.
- Project-specific demand planning and supply-gap detection.
- Cross-system exports (CSV/Excel/PDF/API jobs).

6) Visual Walkthrough

All screenshots below use representative sample data to illustrate expected enterprise workflows.

Login and Session



Role-aware login gates technician vs PI/admin actions.

Generate + Print Cards (Complete Card)

Murisphere Cage Card (Rendered Preview)

Designed for fast room-floor scanning, complete husbandry context, and compliance readiness

Cage: F1-L03-C0215

Group Owner: Dr. Meosha Hudson

Group Name: Hudson Lab

Projects: L03-PRJ-01, L03-PRJ-05

Location: Room 1, Breeding Rack 1, Slot 6A

Protocol: IACUC-2026-0312

Description: PdxCre longitudinal cohort

Breeding: Pairing 35 (Timed Mating)

Protocol Expires: 12/31/2026

Population: M2 / F3 / T5

Scan with phone camera

ID	Sex	DOB	Genotype	Status
258	M	10/10/2024	PdxCre Tg/Azelia	Breeder
1L	F	10/28/2024	PdxCre +/-tg	Breeder
1L1R	F	10/28/2024	PdxCre +/+	Breeder

Litters

#	DOB	Born	Survived	M/F
1	11/12/2024	3	3	2/1

#	DOB	Born	Survived	M/F
2	01/08/2025	5	5	1/4

Complete cage card includes owner/group, protocol, project, animals, litters, and scan-ready QR.

Phone Scan and Quick Edit

Scan/Edit Workflow

1. Enter QR token or scan link

Open Cage

C-A1-0243 (lookup 132ms)

C57BL/6J | WT/WT | Breeding

Room A1 / Rack R3

Male Count Female Count

Save Changes

2. Phone Camera Scan

Open in Browser

Tap banner to open /scan/tok_a1_0243...

Vectorized screenshot: Scan token lookup and edit confirmation

Browser opens cage record directly after phone camera scan.

Cage Alerts and Density View (Sample)

Cage Management - Alert Visualization

Active alerts: 36 (high 9, medium 18). Highlighted cages require immediate review.

ID	Cage	Alert	Status	M/F	Room/Rack
181	F1-L03-C0112	HIGH 3	Breeding	1 / 0	F1-Room-02 / R02-08
264	F1-L03-C0244	MEDIUM 2	Timed Mating	0 / 2	F1-Room-03 / R03-02
309	F1-L03-C0298	LOW 1	Holding	2 / 2	F1-Room-03 / R03-09

Room Density (Sample)

Room	Density
R1	1
R2	2
R3	3
R4	2
R5	3

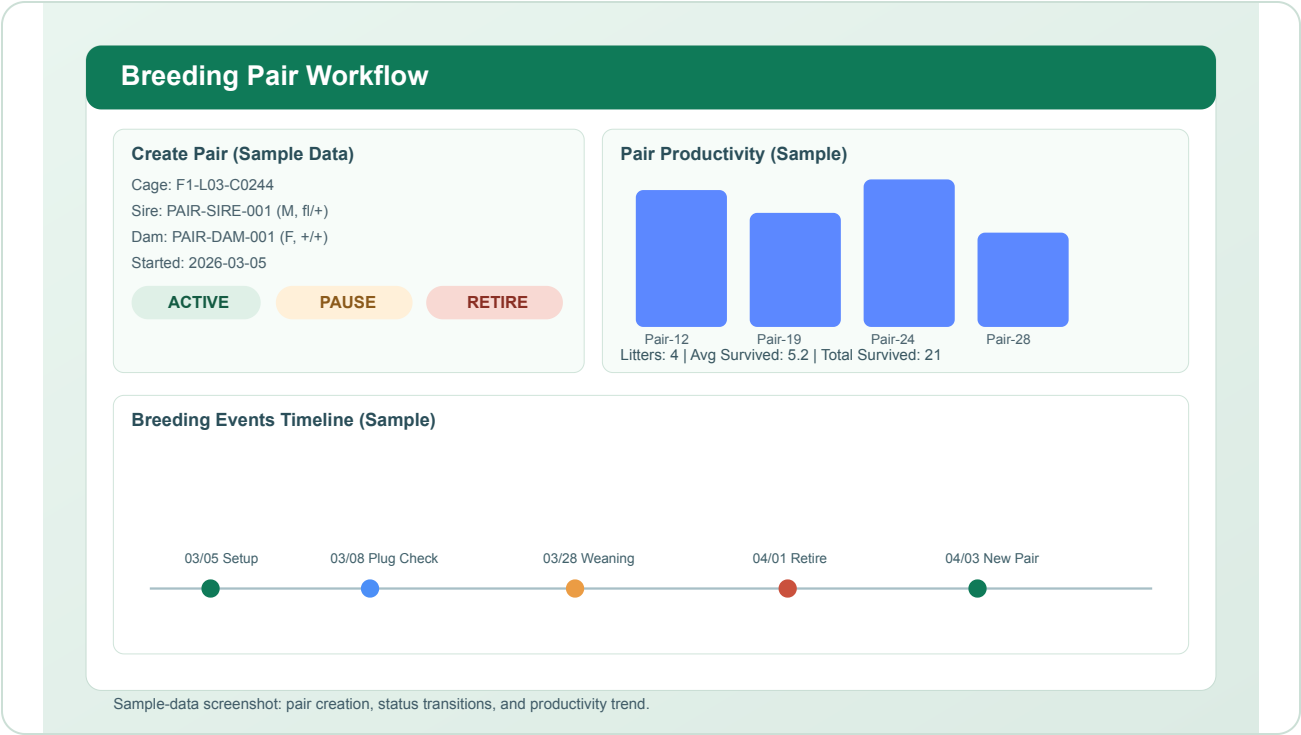
Status Mix (Sample)

Status	Count
Breeding	312
Holding	198
Timed Mating	102

Sample-data screenshot: alert-highlighted cages and at-a-glance room/status visuals.

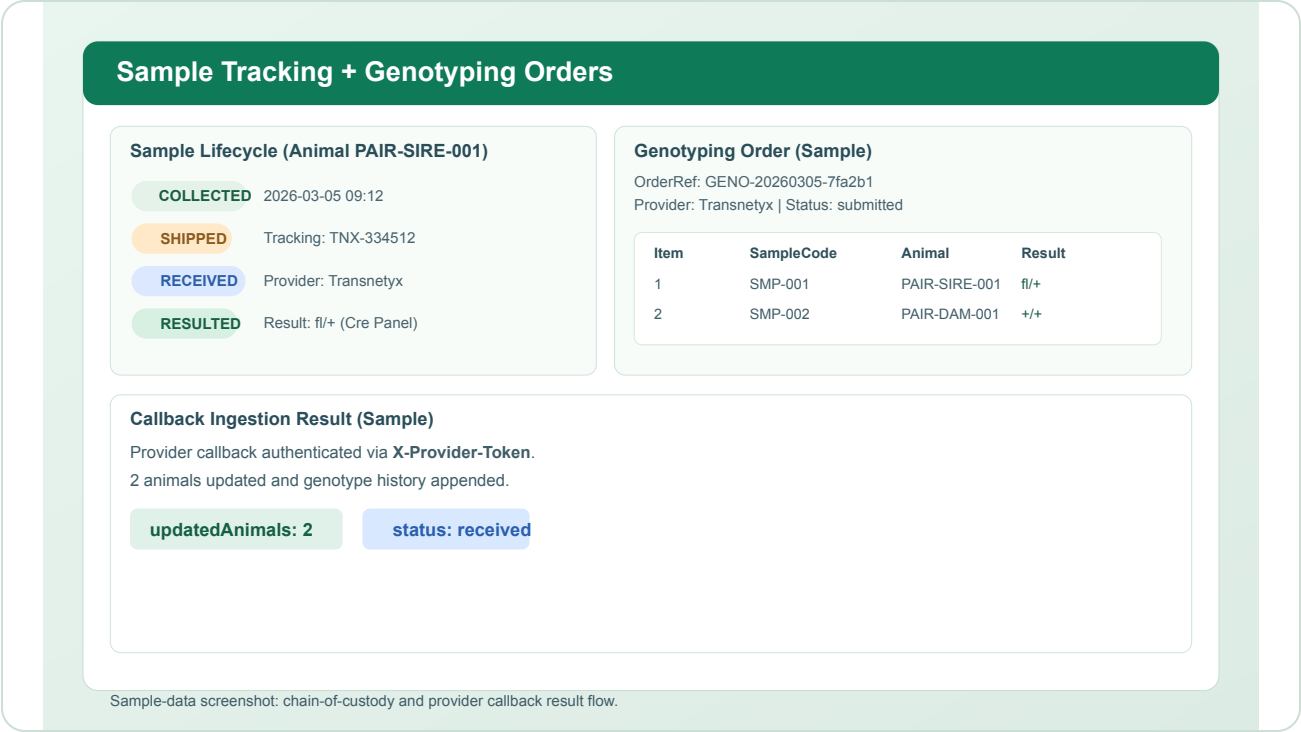
Alert-colored cages and density visuals prioritize technician route order.

Breeding Pair Management (Sample)



Pair lifecycle actions and productivity trend for breeding decisions.

Samples and Genotyping Orders (Sample)



Chain-of-custody plus provider callback result ingestion.

Recommendation Decisions (Sample)

Recommendation Lifecycle

Generated Recommendations (Sample)
Open: 14 | Accepted: 9 | Adjusted: 3 | Ignored: 2 | Completed: 7

Rec #381 - consolidate_cage

Cage F1-L03-C0112 has low density (1 animal) and compatible strain/genotype group nearby.
Suggested target: F1-L03-C0120 | Estimated cage reduction: 1

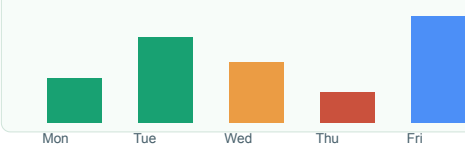
ACCEPT

ADJUST

IGNORE

COMPLETE

Decision Outcome Trend (Sample)



Day	Count	Color
Mon	1	Green
Tue	2	Green
Wed	1	Orange
Thu	1	Red
Fri	3	Blue

Sample-data screenshot: decisionable recommendations with measurable outcomes.

Action Notes (Sample)

Adjusted: "Delay consolidation until post-wean on 2026-03-28"
Accepted by: Admin User at 2026-03-05 14:09
Expected outcome: reduce 12 cages this month.

Accept/adjust/ignore/complete workflow with measurable outcomes.

Planner Scenario Evaluation (Sample)

Planner Scenario and Risk Evaluation

Scenario: Q2 Demand Boost

Needed by: 2026-09-01
Target animals: 1,200
Max new cages: 180
Projects attached: 5

Evaluate

Evaluation Output (Sample)

Projected deficit: 286

Risk level: HIGH

Estimated litters: 72
Estimated cages needed: 58
Recommendation: Increase breeder pairs by 18 over 4 weeks.

Project Demand Allocation (Sample)

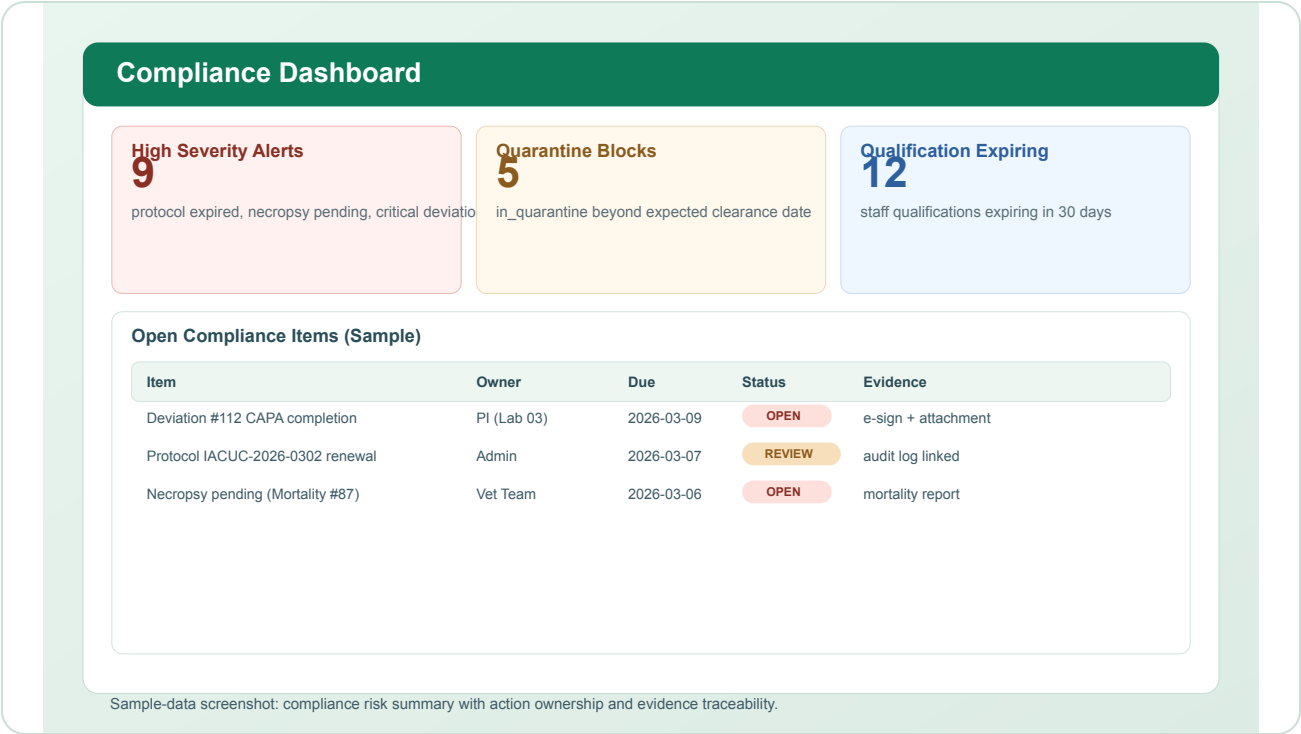


Project ID	Demand
L03-PRJ-02 240	240
L03-PRJ-04 188	188
L03-PRJ-05 272	272
L03-PRJ-01 152	152
L03-PRJ-03 198	198

Sample-data screenshot: planner scenario, risk scoring, and project-level demand profile.

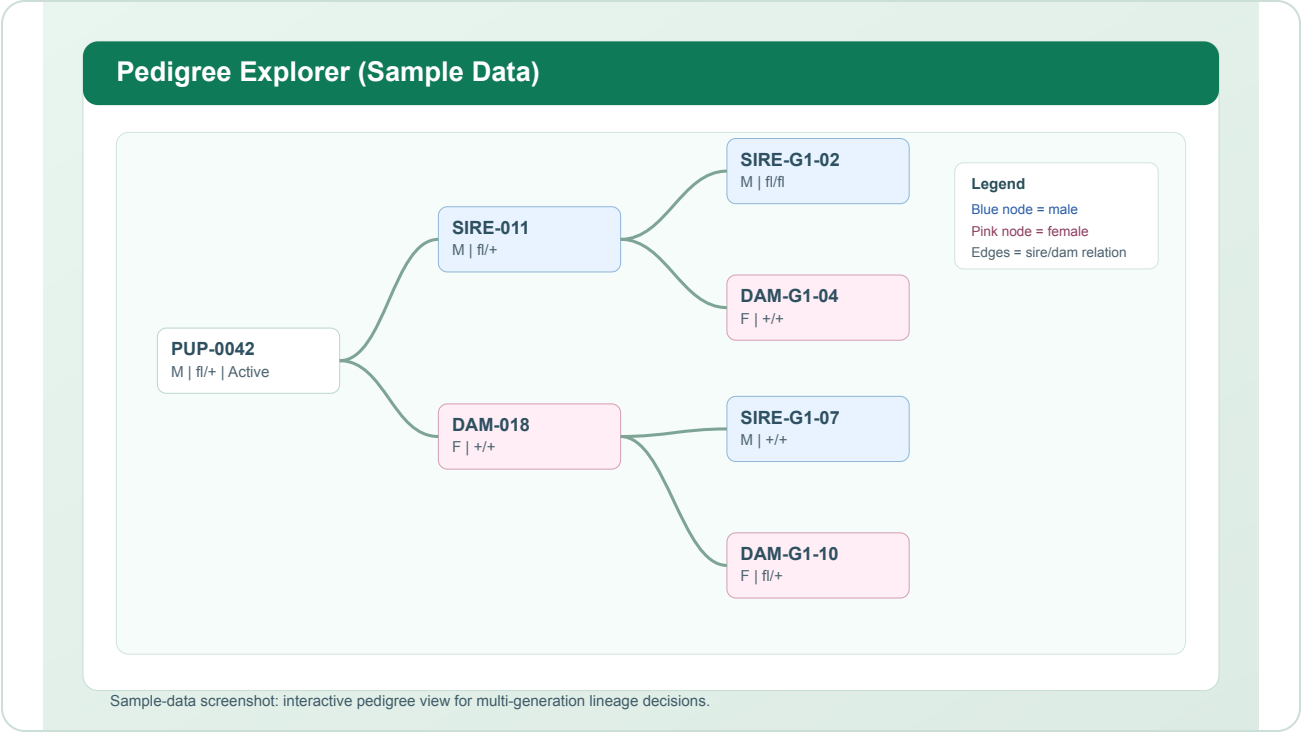
Demand vs supply gap, projected deficit, and risk level by scenario.

Compliance Dashboard (Sample)



Regulatory risk summary with owner, due date, and evidence traceability.

Pedigree Explorer (Sample)



Multi-generation lineage view for breeder planning and genotype interpretation.

7) End-of-Shift / End-of-Day Checklists

Technician End-of-Shift

1. Ensure queued offline writes are fully synced.
2. Confirm all health findings and treatments are documented.
3. Verify mortality/necropsy and euthanasia entries are complete.
4. Close open census/round sessions.

Manager End-of-Day

1. Clear or assign owners for all **high-severity** alerts.
2. Review unresolved deviations and quarantine blocks.
3. Approve/adjust recommendations and planner risk items.
4. Export daily summary for audit and project stakeholders.

8) Troubleshooting and Escalation

- **No camera scan prompt:** improve lighting and flatten printed card.
- **Camera reads barcode value only:** scan the QR square, not the CODE128 bar pattern.
- **Wrong page opens:** correct `Scan Base URL` ; never use localhost for printed cards.
- **QR blank or broken:** verify endpoint `/api/assets/qrcode.png?v=test` .
- **Protocol hard-stop:** reassign valid protocol before edits.
- **Sample or callback mismatch:** validate sample code uniqueness and provider callback token.
- **Planner risk high:** open recommendation workflow and assign corrective action owner.